

From the Least to the Most: Building a Plug-and-Play Visual Reasoner via Data Synthesis

Anonymous ACL submission

Abstract

We explore multi-step reasoning in vision-language models (VLMs). The problem is challenging, as reasoning data consisting of multiple steps of visual and language processing are barely available. To overcome the challenge, we first introduce a least-to-most visual reasoning paradigm, which interleaves steps of decomposing a question into sub-questions and invoking external tools for resolving sub-questions. Based on the paradigm, we further propose a novel data synthesis approach that can automatically create questions and multi-step reasoning paths for an image in a bottom-up manner. Our approach divides the complex synthesis task into a few simple sub-tasks, and (almost entirely) relies on open-sourced models to accomplish the sub-tasks. Therefore, the entire synthesis process is reproducible and cost-efficient, and the synthesized data is quality guaranteed. With the approach, we construct 50k visual reasoning examples. Then, we develop a visual reasoner through supervised fine-tuning, which is capable of generally enhancing the reasoning abilities of a wide range of existing VLMs in a plug-and-play fashion. Extensive experiments indicate that the visual reasoner can consistently and significantly improve four VLMs on four VQA benchmarks.

1 Introduction

Large language models (LLMs) (Brown et al., 2020; Achiam et al., 2023) have demonstrated remarkable performance across various tasks in natural language processing (NLP). Encouraged by the success, the artificial intelligence community is now enthusiastically exploring ways to enable LLMs to process information from modalities beyond language, leading to a recent surge in the development of large multimodal models (LMMs) (Alayrac et al., 2022; Liu et al., 2024b; Team et al., 2023; Dai et al., 2024; Bai et al., 2023; Ormazabal et al., 2024). By adopting a pre-training

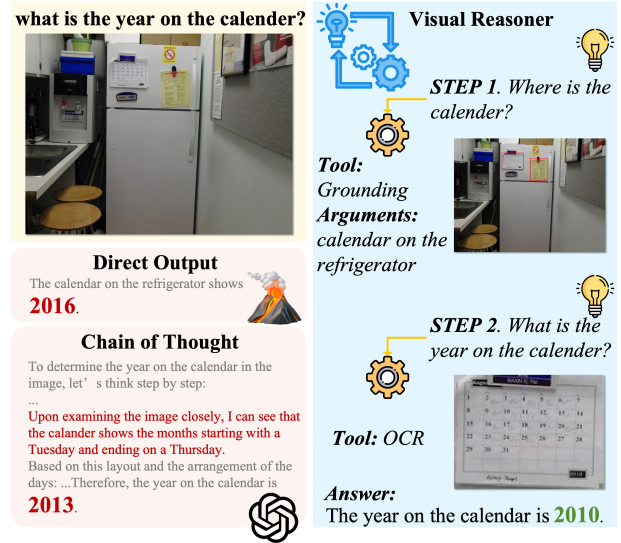


Figure 1: **Top Left:** An example from TextVQA (Singh et al., 2019) with “2010” as the ground truth. **Middle Left:** Response from LLaVA-NeXT-13B (Liu et al., 2024a). **Bottom Left:** Response from GPT-4o with the prompt “{question} please think step by step and answer the question”. **Right:** Response given by the proposed method, which is also the only correct answer.

and instruction tuning paradigm, representations of multiple modalities are effectively fused in deep architectures, bringing substantial advancements in tasks, such as image captioning (Young et al., 2014), visual question answering (VQA) (Goyal et al., 2017), and optical character recognition (OCR) (Mishra et al., 2019), among others.

In this work, we explore vision language models (VLMs) as a typical example of LMMs. Despite the rapid progress, state-of-the-art VLMs still face challenges in reasoning over visual content. As exemplified in Figure 1, intuitively, the question can be correctly solved following a “least-to-most” paradigm (Zhou et al., 2022), in which the question is decomposed into a series of sub-questions, and an answer is deduced by resolving the sub-questions step by step. The problem is that existing

VLMs are inept at performing such multi-step reasoning because (1) Multi-step reasoning paths, like the one shown in Figure 1 (right), are rarely included in the training data (Dai et al., 2024)¹. The VLMs have few opportunities to develop the reasoning capability from the subsequent post-training. (2) Different from reasoning over text, solving questions in a vision-language context may require manipulating the input image (e.g., marking a specific area) and deducting the next steps from both textual and visual intermediate results. The requirement, however, is difficult to accomplish for most VLMs, whether open-sourced or proprietary.

We present *least-to-most visual reasoning*, a general paradigm to guide VLMs to decompose a given question into sub-questions and invoke tools to resolve each sub-question for handling diverse visual reasoning tasks. While there have been extensive studies for LLMs regarding chain-of-thought reasoning (Wei et al., 2022b; Yao et al., 2022, 2024; Wang et al., 2022) and tool-invoking (Schick et al., 2024; Qin et al., 2023b), the techniques are less explored in the context of VLMs. Since data scarcity is a major obstacle, we propose a novel data synthesis approach, dubbed *least-to-most synthesis*, to automatically generate a “(question, reasoning path)” tuple for a given image in a bottom-up manner. Specifically, the pipeline of *least-to-most synthesis* consists of four steps: (1) *Entity Recognition*: recognizing all entities in an image; (2) *Node Construction*: constructing three types of nodes, each aggregating an image with a few entities and some textual features; (3) *Reasoning Process Synthesis*: synthesizing a reasoning path from a sampled chain of nodes. Based on the nodes, the reasoning path is formed by connecting a sequence of sub-questions and tool arguments generated by an LLM; and (4) *Question Synthesis*: generating the main question by recursively combining the sub-questions in the reasoning path through an LLM. Our approach (almost entirely) relies on open-sourced models. Therefore, it offers several advantages, including cost-efficiency, reproducibility, and ensured data quality, over the common practice where data are obtained by querying powerful proprietary LLMs like GPT-4V (Qi et al., 2024; Li et al., 2024).

Based on *least-to-most synthesis*, we build a

large scale **V**isual **RE**asoning dataset (VIREO) with 50k examples, and tailor LLaVA-1.5-7B (Liu et al., 2023a) as a visual reasoner through supervised fine-tuning on VIREO. The reasoner can be generally applied to off-the-shelf VLMs in a plug-and-play fashion to enhance their reasoning capabilities. We conduct experiments with four representative VLMs as showcases. Evaluation results across four VQA benchmarks indicate that the reasoner can consistently improve all VLMs over all tasks, with absolute performance gains ranging from 0.71% to 39%.

Our contributions are three-fold:

I. We introduce the *least-to-most visual reasoning* paradigm to synergize question-decomposition and tool-invoking in VLMs for solving complex vision-language tasks.

II. We propose *least-to-most synthesis*, a reproducible, cost-efficient, and data quality-assured algorithm for automatically creating multi-step visual reasoning data (almost) using open-source models.

III. We use *least-to-most synthesis* to construct the VIREO dataset of 50k examples for fine-tuning a reasoner model. Extensive experiments illustrate that the reasoner can consistently and significantly enhance existing VLMs in a plug-and-play fashion across five VQA benchmarks².

2 Related Works

2.1 Vision-Language Models

Building LMMs aims to enable foundation models to seamlessly handle multimodal signals, such as language, vision, and audio. Among the efforts, significant attention has been focused on jointly modeling vision and language, known as VLMs. Recent work on VLMs can be broadly categorized into two groups, according to how visual information is incorporated into the models. The first line integrates visual information into LLMs via a vision-language connector. For example, LLaVA series (Liu et al., 2023b,a) exploit a linear transformation or an MLP to transform outputs from a vision encoder into inputs of a language model. BLIP-2 (Li et al., 2023a) and InstructBLIP (Dai et al., 2024) rely on Query Transformers to achieve vision-language alignment. Similarly, Qwen-VL (Bai et al., 2023) and mPLUG-Owl (Ye et al., 2023) use learnable tokens to take visual information into account. CogVLM (Wang et al., 2023a) maps vision embedding to the space of word embedding by an MLP adapter, and then

¹A few datasets, such as A-OKVQA (Schwenk et al., 2022), VCR (Zellers et al., 2019), and ScienceQA (Lu et al., 2022), contain rationales. However, the rationales merely provide explanations for the answers and thus differ significantly from the multi-step reasoning data we study in the work.

²All the code and data will be published later.

enables deep vision-language feature alignment via a visual expert module. The second group strives to train VLMs natively from image tokens and textual tokens. Among the representative models, Flamingo (Alayrac et al., 2022) inserts gated cross-attention dense blocks into a pre-trained LM. BEiT-3 (Huang et al., 2024) utilizes Multiway Transformers as the backbone to encode various modalities. KOSMOS-1 (Huang et al., 2024) and KOSMOS-2 (Peng et al., 2023) interleave image tokens with textual tokens in the input sequence through a designed format. Gemini family (Team et al., 2023; Reid et al., 2024) take multimodal signals as input and can natively generate images using discrete image tokens. Instead of building a new VLM, we target enhancing the multi-step reasoning ability of existing VLMs. Our data synthesis approach enables us to develop a visual reasoner that can be generally applied to various VLMs, leading to consistent improvements over various VQA tasks.

2.2 Reasoning and Tool Use

Reasoning is an important emergent ability of LLMs (Wei et al., 2022a). With appropriate exemplars or prompts, LLMs can demonstrate “chain-of-thought” (CoT) behavior and solve problems through multi-step reasoning (Wei et al., 2022b; Kojima et al., 2022). Encouraged by the observation, significant efforts have been made to improve the reasoning capabilities of LLMs. For example, Zhou et al. (2022) propose least-to-most prompting for complex reasoning; Yao et al. (2022) extend the ability of LLMs by interleaving reasoning and acting; Yao et al. (2024) introduce tree-of-thought (ToT) as an extension of CoT; and Besta et al. (2024) further extend the idea of ToT by considering a graph structure. In addition to innovations in methodology, reasoning abilities have also proven effective in various applications, such as table understanding (Wang et al., 2023c), math problem solving (Wei et al., 2022b), question answering (Guan et al., 2024), and decision making (Yao et al., 2022). Very recently, the research community begins to investigate the problem in multimodal models (Qi et al., 2024; Yang et al., 2023; Wang et al., 2024). Different from existing work, we focus on data synthesis with open-sourced models, and develop a plug-an-play visual reasoner.

Our work also relates to the efforts on facilitating LLMs to leverage tools (Qin et al., 2023b; Schick et al., 2024; Shen et al., 2024; Qin et al., 2023a). The difference is that our tools are spe-

cially selected for visual processing, with the goal of enhancing the reasoning capabilities of VLMs.

3 Method

We elaborate on our method for multi-step reasoning in VLMs. First, we formalize *least-to-most visual reasoning* that delineates how a visual reasoner solves a complex problem according to an image (§3.1). Then, we present details of *least-to-most synthesis* by which a VLM can be tuned as the visual reasoner to perform reasoning (§3.2).

3.1 Least-to-Most Visual Reasoning

We formalize the *least-to-most visual reasoning* paradigm as follows: Given an image I and a question Q , a visual reasoner $\mathcal{M}_R$ deduces a multi-step reasoning path R , where each step either performs operations on I (e.g., marking an area with a red box) or asks an off-the-shelf VLM $\mathcal{M}$ to conclude a final answer. To this end, we represent R as a chain of invoking tools from a pre-defined pool $\mathcal{T} = \{t^i | i = 1, 2, \dots, T\}$ step by step, where each tool refers to a class of operation on images or the VLM $\mathcal{M}$, as illustrated in Figure 2 (right).

Reasoning Process. At the k -th step, $\mathcal{M}_R$ proposes a sub-question q_k and selects a tool t_k from $\mathcal{T}$ based on an image I_k and previous steps:

$$q_k, t_k = \mathcal{M}_R(I_k, Q, \{q_{<k}\}), \quad (1)$$

where t_k is a textual description specifying the invoked tool and the corresponding arguments. Then, the answer to q_k is deduced by t_k , denoted as r_k , based on which I_{k+1} is obtained as follows:

$$I_{k+1} = \begin{cases} r_k, & \text{if } r_k \text{ is an image,} \\ I_k, & \text{otherwise.} \end{cases} \quad (2)$$

Particularly, we set I_1 to I . If I_k includes a red box to mark some area smaller than a threshold α , and t_k intends to infer information from I_k (including the OCR and Answer tools in Table 1), we automatically crop this area from original I_k and enlarge it to the same size as I_k . The above process iterates until t_k refers to the VLM $\mathcal{M}$, in which case we define the final answer $A = r_k$ and terminate the reasoning process. In summary, we formally denote $R = [(I_k, q_k, t_k)]_{k=1}^K$,

Tools. Following human experience, we define four tools, each targeting a class of atomic problems and outputs either a modified image or a piece of text. Tab. 1 describes details.

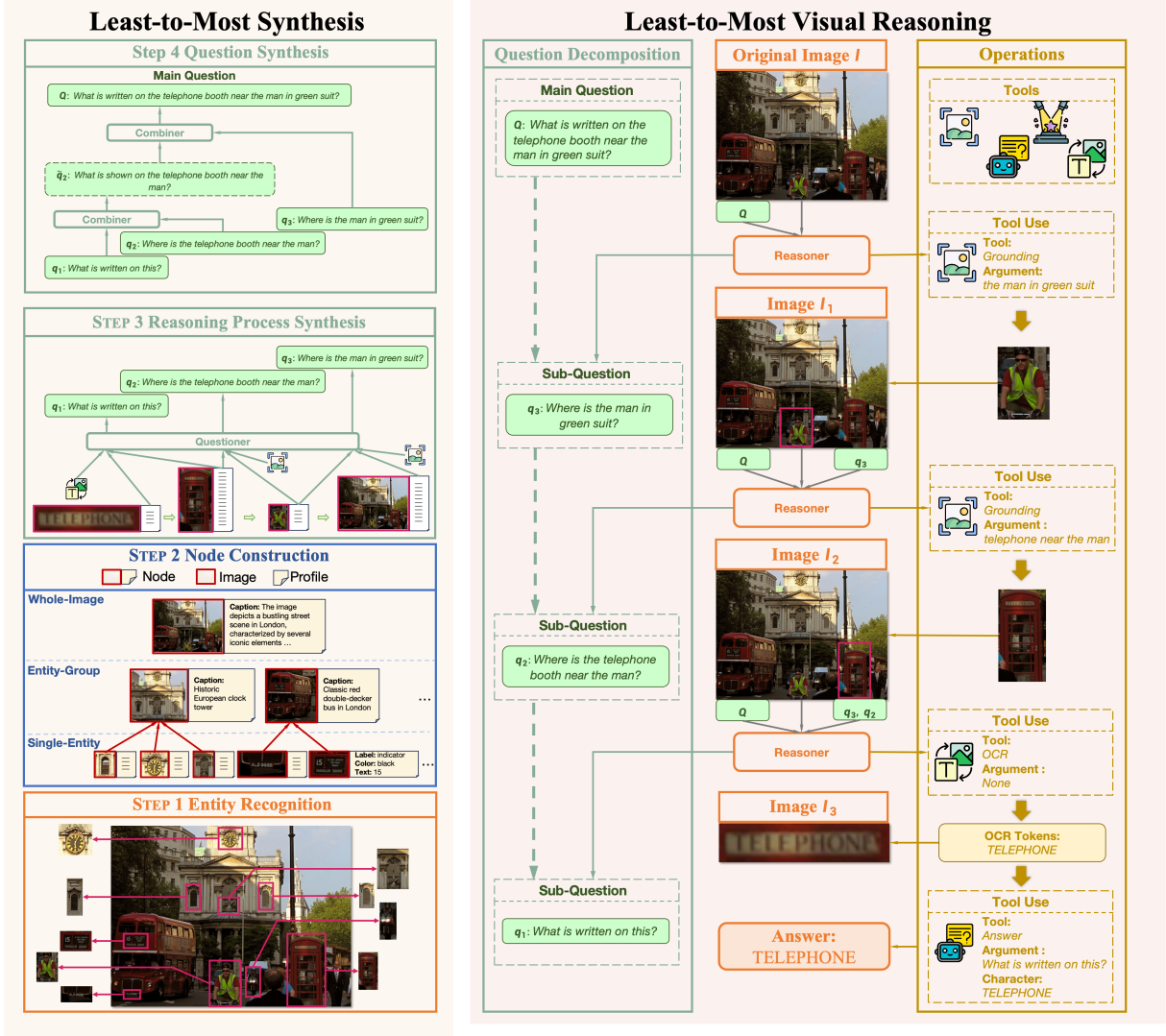


Figure 2: **Left:** the pipeline of *least-to-most synthesis*. **Right:** the process of *least-to-most visual reasoning*.

Specifically, we implement Grounding and Highlight using GroundingDino (Liu et al., 2023c), OCR using PaddleOCR³, and Answer using the VLM $\mathcal{M}$. The reasoner is trained to invoke these tools to dive into the details of the given image. By varying $\mathcal{M}$ in Answer, the reasoner can adapt to different VLMs in a plug-and-play fashion, and significantly enhance their performance across a wide range of tasks, as will be shown in §4.

3.2 Least-to-Most Synthesis

To overcome the data barrier, a common practice is to feed image-question pairs to powerful proprietary LMMs like GPT-4V, and gather the outputs as a dataset (Qi et al., 2024). The top-down approach, however, suffers from several issues: (1) Even powerful proprietary models like GPT-4V still struggle

to perform reliable reasoning (Wu et al., 2023), implying that the quality of data obtained in this way is not guaranteed. (2) Utilizing proprietary models incurs high costs, hindering the scalability of the synthesized data. (3) It is difficult for others to reproduce the method since the behavior of proprietary models can vary over time.

In contrast to the top-down approach, we propose a bottom-up pipeline that can synthesize multi-step visual reasoning data using (almost entirely) open-sourced models while ensuring the quality of the synthesized data. The workflow begins with an image, gradually generates sub-questions associated with tools and intermediate results, and ultimately synthesizes a question based on the image and the reasoning path. Specifically, the pipeline comprises four steps: *entity recognition*, *node construction*, *reasoning path synthesis*, and *question*

³<https://github.com/PaddlePaddle/PaddleOCR>

Tool	Argument		Output	Purpose
 Grounding	Image 	Target Entity The woman in a purple ski suit.	Image 	Frame an area from the Image corresponding to the Target Entity
 Highlight	Image 	Target Entity Pink Cookie	Image 	Highlight all entities corresponding to the Target Entity in the Image
 OCR	Image 		List of Text ["CURLY DICK RD", "Tarana 13", "Oberon 39"]	Recognize all pieces of text from the Image
 Answer	Image 	Question What is the next stop? Character ["NEXT STOP", "BWI AIRPORT"]	Answer As shown in the image, the next stop is "BWI STOP".	Answer the Question based on the Image and recognized Character . And Character can be empty.

Table 1: Tools used in our reasoner.

synthesis, where each node aggregates a focused (sub-)image and relevant text-form information. Figure 2 (Left) illustrates the construction process.

Entity Recognition. We employ Deformable DETR (Zhu et al., 2020) to recognize entities in an image, which can identify 1,203 types of entities. In practice, we discard those entities with confidence scores ≤ 0.5 .

Node Construction. Given an image, we automatically construct nodes based on the recognized entities. Each node is represented as a pair of an image and a textual profile, where the image is extracted from the given one, and the profile is an attribute-value dictionary about the image. By converting the image into a textual profile, data synthesis can eliminate the reliance on visual signals, allowing the use of more advanced LLMs instead of VLMs in subsequent processes. As indicated in Figure 2, we define three types of nodes spanning various granularities: **(1) Single-Entity Node.** The image for such a node refers to one recognized entity. We utilize specialized tools to extract accurate and fine-grained attributes from multiple dimensions (e.g., color) to form the profile. More details are presented in Appendix A. **(2) Entity-Group Node.** The image of an entity-group node is the aggregation of multiple recognized entities that are close to each other. We employ BLIP⁴ to caption the image as the corresponding profile of the node. Each caption contains about 10-20 tokens, illustrating the inter-entity relations in detail. **(3) Whole-**

Image Node. This type of node corresponds to the whole image given in advance. We use LLaVA⁵ to caption the image in detail as its profile, typically exceeding 200 tokens in length, which offers comprehensive but coarse-grained information.

Reasoning Process Synthesis. We sample a chain of M nodes from the constructed node set, denoted as $(N_1, N_2, \dots, N_M)$, which will be connected in turn to form the reasoning process. We craft elaborate rules to ensure nodes can be reasonably connected and the last node N_M is a whole-image node, as detailed in Appendix A. For every two adjacent nodes N_m and N_{m+1} in the chain ($m \leq M - 1$), based on their profiles and a sampled tool t_m , we exploit an LLM as a Questioner to synthesize one sub-question q_m to ask about a certain attribute of the head node N_m conditioned on the tail node N_{m+1} :

$$q_m, \hat{t}_m = \text{Questioner}(N_m^P, N_{m+1}^P, t_m), \quad (3)$$

where N_m^P and N_{m+1}^P refer to the profiles of N_m and N_{m+1} , respectively, $\hat{t}_m$ is the argument of the specified tool t_m to solve q_m . Iterating the process w.r.t. m , we obtain $M - 1$ sub-questions and synthesize the entire reasoning process R as $[(I_{M-1}, q_{M-1}, \hat{t}_{M-1}), \dots, (I_1, q_1, \hat{t}_1)]$, where I_m is the image of N_m ($m = 1, \dots, M - 1$).

Question Synthesis. We generate the main question Q by recursively combining the sub-questions

⁴<https://huggingface.co/Salesforce/blip-image-captioning-large>

⁵<https://huggingface.co/llava-hf/llava-v1.6-vicuna-13b-hf>

using another LLM as a Combiner:

$$Q = \tilde{q}_{M-1}, \quad (4)$$

$$\tilde{q}_m = \begin{cases} q_1, & m = 1, \\ \text{Combiner}(\tilde{q}_{m-1}, q_m), & m > 1, \end{cases} \quad (5)$$

where $\tilde{q}_m$ is an intermediate result for the m -th step. Note that $\{\tilde{q}_m\}_{m=2}^{M-2}$ are excluded from R , as they are just used for synthesis of Q .

We implement Questioner and Combiner by fine-tuning LLaMA-3-8B-Instruct⁶. Due to limited resources, we obtain the training data by querying GPT-4 (Wang et al., 2023b) using a few high-quality demonstrations as seeds following the self-instruct pipeline (Wang et al., 2023b). Since the two tasks are relatively simple, querying GPT-4 only 10k times is enough to achieve satisfactory performance⁷. More details are in Appendix B.

Compared to the top-down approach, our *least-to-most synthesis* is reproducible and significantly more cost-efficient. Moreover, every step in the pipeline is atomic, ensuring the performance of the open-sourced models on these simple tasks. Therefore, the synthesized data is guaranteed to be of high quality, as will be demonstrated in §4.

4 Experiments

To assess the efficacy of the proposed method, we fine-tune an LLaVA-1.5-7B⁸ as the Reasoner, and plug the Reasoner into four representative VLMs with various sizes and conduct experiments on four standard VQA benchmarks. In the implementation, we obtain 10k training examples for Question and Combiner, respectively, by querying GPT-4. Subsequently, we synthesize a visual reasoning dataset (VIREO) with 50k examples following the least-to-most synthesis approach, and perform instruction tuning on VIREO to obtain the Reasoner. More implementation details are presented in Appendix B.

4.1 Models for the Answer Tool

To comprehensively illustrate the versatility of our Reasoner, we employ several VLMs with different sizes and architectures as the Answer tool, including (1) **BLIP-2** (Li et al., 2023a): It utilize a Q-Former module to integrate visual and textual in-

formation. We use the 2.7B version⁹, which is the smallest model in our experiments. (2) **Instruct-BLIP** (Dai et al., 2024): It expands BLIP-2 by incorporating instruction prompts into the Q-Former module. We use the 7B¹⁰ and 13B¹¹ versions in our experiments. (3) **LLaVA** (Liu et al., 2024a): It hinges on a basic projection layer to align image and text representations. We use the 13B version¹².

4.2 Evaluation Datasets

We conduct experiments on the following datasets: (1) **GQA** (Hudson and Manning, 2019): It is a VQA dataset constructed from knowledge graphs, primarily focusing on inter-entity attribute relationships. (2) **TextVQA** (Singh et al., 2019) and **ST-VQA** (Biten et al., 2019): The two datasets include textual information within images, used to evaluate the capability to understand text in pictorial form. (3) **TallyQA** (Acharya et al., 2019): It is widely used to assess the counting ability, divided into a simple subset and a complex subset. The complex subset of TallyQA involves more fine-grained attributes of the entities in the images than the simple subset. In our experiments, we denote these subsets as Tally-S and Tally-C, respectively.

Since the VLMs used for the Answer tool are general-purpose generative models and not fine-tuned on the evaluation datasets, they may produce correct answers but include additional information (e.g., “The answer is”) beyond the ground truth provided by the datasets. Therefore, we use Exact Matching (EM) as the metric only for TallyQA. For GQA and TextVQA, we use answer recall, i.e., whether the output includes the ground truth, for evaluation. For ST-VQA, we submit results to its official website.

4.3 Main Results

Table 2 reports evaluation results. We find that (1) **The Reasoner consistently improves the performance of all VLMs across all datasets.** By decomposing questions and invoking specialized tools, the Reasoner can improve various off-the-shelf VLMs in a plug-and-play fashion, suggesting its strong generalization ability. (2) **The Reasoner**

⁶<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

⁷We will publish Questioner and Combiner models later.

⁸<https://huggingface.co/llava-hf/llava-1.5-7b-hf>

⁹<https://huggingface.co/Salesforce/blip2-opt-2.7b>

¹⁰<https://huggingface.co/Salesforce/instructblip-vicuna-7b>

¹¹<https://huggingface.co/Salesforce/instructblip-vicuna-13b>

¹²<https://huggingface.co/llava-hf/llava-v1.6-vicuna-13b-hf>

Model	Size	GQA	TextVQA	ST-VQA	TallyQA-S	TallyQA-C
BLIP2	2.7B	40.85	32.65	18.89	20.93	29.00
+ Reasoner	-	42.02 (+1.17)	36.94 (+4.29)	19.60 (+0.71)	56.38 (+35.45)	39.38 (+10.38)
InstructBLIP	7B	51.65	38.46	24.85	67.05	38.84
+ Reasoner	-	53.60 (+1.95)	43.42 (+4.96)	26.36 (+1.51)	74.99 (+7.94)	57.78 (+18.94)
InstructBLIP	13B	52.72	37.61	22.21	58.38	23.46
+ Reasoner	-	54.24 (+1.48)	42.03 (+4.42)	23.72 (+1.51)	74.98 (+16.60)	59.73 (+36.27)
LLaVA	13B	57.02	57.04	-	70.29	29.69
+ Reasoner	-	60.65 (+3.63)	59.22 (+2.18)	-	80.28 (+9.99)	68.65 (+38.96)

Table 2: Evaluation results with different VLMs as the Answer tools. **Size** refers to the number of parameters of the language model in each VLM. We do not provide the result of LLaVA on ST-VQA because its output always includes abundant information, leading to nonsensical low scores using the official evaluation website.

helps better capture complex inter-entity relations. The Reasoner brings clear improvements on GQA, which involves diverse relations among entities in the images. The improvements could result from the least-to-most synthesis algorithm in which such relations are implicitly modeled. The improvement for LLaVA (3.63%) is more substantial than other weaker VLMs (1.17%-1.95%). It is possibly because stronger VLMs may better realize their potential with the Reasoner considerably alleviating the difficulty in locating the target entity. **(3) The Reasoner boosts the understanding of words in images.** The OCR tool empowers the Reasoner to effectively enhance the performance on both TextVQA and ST-VQA. On TextVQA, we notice that the enhancement is less significant on LLaVA than on weaker VLMs, possibly because LLaVA is less dependent on external tools for understanding the words in images. **(4) The Reasoner improves the counting performance by a large margin.** The benefit of the Reasoner is particularly significant on TallyQA. Without the Reasoner, the base VLMs are sensitive to irrelevant information in the images and thus easily miscount. By utilizing the Highlight tool, the Reasoner can reliably mitigate the impact of such irrelevant information.

4.4 Quality Assessment of VIREO

We conduct a human study to investigate the quality of our synthesized VIREO dataset. Specifically, we randomly sample 200 examples from VIREO, and recruit 3 graduate students majoring in NLP to manually check the correctness of the synthesized question and reasoning process for each instance. The check focuses on the correctness in three aspects: **(1) Sub-Question.** The sub-question must be accurately faithful to the profile of the tail node and can be resolved using the specified tool. **(2)**

Argument. Based on the current sub-question, the annotators need to assess whether using the given argument to invoke the tool can yield the expected output. **(3) Main Question.** The annotators judge whether the main question covers all sub-questions and whether it can be decomposed into sub-questions in the correct order. Each example receives 3 labels from each of the three annotators on each aspect. Results show that the three evaluators consistently approve that all examples are correct in all aspects¹³, suggesting the highly guaranteed quality of the data produced by the proposed synthesis approach.

5 Discussions

Although the main results affirmed the high quality of VIREO and the general benefits of the Reasoner to enhance VLMs on multiple benchmarks, we are still curious about the following research questions: **(1) RQ1:** How does the capability of the Reasoner vary with respect to the size of instructions for fine-tuning? **(2) RQ2:** Will the Reasoner generate negative impact on visual tasks other than VQA? **(3) RQ3:** What are the underlying causes of the Reasoner’s errors on the benchmarks?

5.1 RQ1: Regarding Influence of Data Size

Figure 3 presents the model performance when we vary the size of VIREO. Overall, the gain in performance gradually increases with the number of training examples. However, we also notice that the marginal benefit is diminishing, suggesting that more diverse data synthesis approaches besides ours are needed to further break through the upper limit in future work.

¹³This might be surprising, but we double-checked the evaluation and confirmed the results.

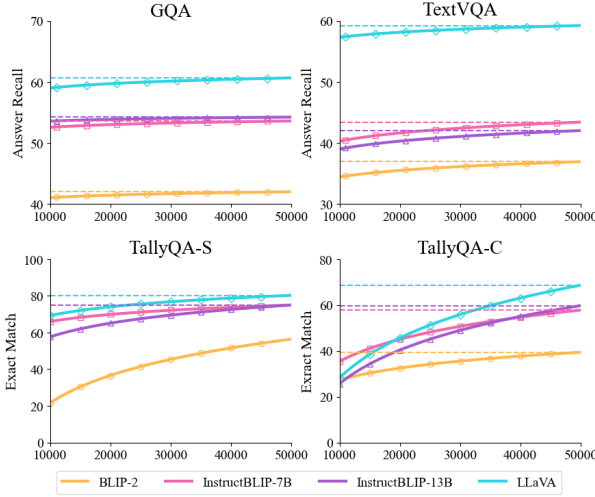


Figure 3: The performance of the Reasoner varies with the size of VIREO. The dashed lines indicate the performance of corresponding models using 50k training examples.

Model	Size	MMMU	POPE
InstructBLIP	7B	22.43	84.79
+ Reasoner	-	22.50 (+0.07)	84.79 (+0.00)
InstructBLIP	13B	18.75	80.42
+ Reasoner	-	18.23 (-0.52)	80.42 (+0.00)

Table 3: Evaluation results on other visual tasks.

5.2 RQ2: Regarding Other Visual Tasks

To investigate whether introducing a Reasoner will hurt the performance on other visual tasks, we conduct experiments on MMMU (Yue et al., 2024) and POPE (Li et al., 2023b) as examples. The MMMU benchmark aggregates massive multi-discipline tasks necessitating abundant knowledge to address, which is beyond the scope of our Reasoner. POPE is a discrimination dataset for hallucination detection which requires discriminating whether a certain coarse-grained object is present in a given image without the need for multi-step reasoning. We choose InstructBLIP-7B and -13B as the base VLMs for discussion.

As shown in Table 3, introducing the Reasoner on MMMU has a minimal impact on the final performance, suggesting that the Reasoner does not influence the expression of VLMs’ inherent knowledge. On the other hand, the presence or absence of the Reasoner on POPE yields consistent results, indicating that additional reasoning over images does not increase the likelihood of the VLMs generating hallucinations.

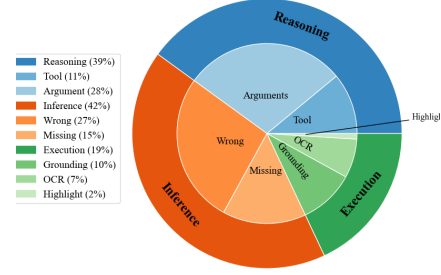


Figure 4: The distribution of different error types.

5.3 RQ3: Regarding Error Analysis

To analyze the error types of the Reasoner, we randomly sample 100 instances where the model prediction is wrong from all evaluation sets. Then, we analyze their error and categorize the errors into three types: **(1) Reasoning:** The Reasoner uses a wrong tool (*Tool*) or generates wrong arguments for the tool (*Arguments*); **(2) Execution:** The *Grounding*, *OCR*, or *Highlight* tool returns wrong execution results; and **(3) Inference:** The *Answer* tool outputs wrong answers (*Wrong*) or irrelevant answers with the question (*Missing*).

As shown in Figure 4, the “Inference” error has the largest proportion (42%). Among these, there are even 15% of instances where the *Answer* tool generates irrelevant answers, indicating the huge room to improve the instruction-following ability of existing VLMs. Additionally, the “Reasoning” error accounts for a significant proportion (39%), and most of the instances (28%) are attributed to wrong arguments. This means VLMs’ tool-invoking capability is far from perfect, yet this ability is crucial for VLMs to interact with the environment. The “Execution” error is less frequent, implying the huge potential of building general and powerful VLMs based on specialized visual tools.

6 Conclusions

We propose a data synthesis approach to enhancing multi-step reasoning capabilities of vision-language models. The approach decomposes the synthesis task into several simple sub-tasks, and finish the sub-tasks (almost) with open-sourced models. Based on the approach, we build a large-scale visual reasoning dataset, and develop a visual reasoner by tuning on the dataset. Evaluation results across four VQA benchmarks indicate that the visual reasoner can generally improve the reasoning capabilities of a number of existing VLMs.

Limitations

We select COCO2014 (Lin et al., 2014) as the source of images for our data synthetic process. However, while COCO2014 is a general-purpose image dataset, we do not guarantee that its data can cover all visual tasks. Additionally, our proposed method demonstrates consistent improvements across four VQA benchmarks, but this does not imply that our method will be effective in all visual datasets and scenarios. Furthermore, during our experiments, we only selected four VLMs as base models, and we cannot ensure that our method is capable of bringing improvements to all models.

Ethical Considerations

Images may contain sensitive information. Using or publishing datasets that include such information could pose potential ethical risks. In our experiments, we strictly control the image sources of our data, utilizing only authorized open-source datasets. Furthermore, all the models we used are publicly accessible and adhere to established requirements.

References

Manoj Acharya, Kushal Kafle, and Christopher Kanan. 2019. Tallyqa: Answering complex counting questions. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 8076–8084.

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. 2022. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736.

Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-vl: A frontier large vision-language model with versatile abilities. *arXiv preprint arXiv:2308.12966*.

Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, et al. 2024. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17682–17690.

Ali Furkan Biten, Ruben Tito, Andres Mafla, Lluís Gomez, Marçal Rusinol, Ernest Valveny, CV Jawahar, and Dimosthenis Karatzas. 2019. Scene text visual question answering. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4291–4301.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. 2024. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in Neural Information Processing Systems*, 36.

Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6904–6913.

Jian Guan, Wei Wu, Zujie Wen, Peng Xu, Hongning Wang, and Minlie Huang. 2024. Amor: A recipe for building adaptable modular knowledge agents through process feedback. *arXiv preprint arXiv:2402.01469*.

Shaohan Huang, Li Dong, Wenhui Wang, Yaru Hao, Saksham Singhal, Shuming Ma, Tengchao Lv, Lei Cui, Owais Khan Mohammed, Barun Patra, et al. 2024. Language is not all you need: Aligning perception with language models. *Advances in Neural Information Processing Systems*, 36.

Drew A Hudson and Christopher D Manning. 2019. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6700–6709.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023a. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR.

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. 2023b. Evaluating object hallucination in large vision-language models. *arXiv preprint arXiv:2305.10355*.

Zejun Li, Ruipu Luo, Jiwen Zhang, Minghui Qiu, and Zhongyu Wei. 2024. Vocot: Unleashing visually

665	grounded multi-step reasoning in large multi-modal	Yujia Qin, Shengding Hu, Yankai Lin, Weize Chen,	718
666	models. <i>arXiv preprint arXiv:2405.16919</i> .	Ning Ding, Ganqu Cui, Zheni Zeng, Yufei Huang,	719
667		Chaojun Xiao, Chi Han, et al. 2023a. Tool	720
668	Tsung-Yi Lin, Michael Maire, Serge Belongie, James	learning with foundation models. <i>arXiv preprint</i>	721
669	Hays, Pietro Perona, Deva Ramanan, Piotr Dollár,	<i>arXiv:2304.08354</i> .	722
670	and C Lawrence Zitnick. 2014. Microsoft coco:		
671	Common objects in context. In <i>Computer Vision–</i>	Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan	723
672	<i>ECCV 2014: 13th European Conference, Zurich,</i>	Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang,	724
673	<i>Switzerland, September 6–12, 2014, Proceedings,</i>	Bill Qian, et al. 2023b. Toolllm: Facilitating large	725
	<i>Part V 13</i> , pages 740–755. Springer.	language models to master 16000+ real-world apis.	726
674		<i>arXiv preprint arXiv:2307.16789</i> .	727
675	Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae		
676	Lee. 2023a. Improved baselines with visual instruc-	Machel Reid, Nikolay Savinov, Denis Teplyashin,	728
	tion tuning.	Dmitry Lepikhin, Timothy Lillicrap, Jean-baptiste	729
677		Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Fi-	730
678	Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan	rat, Julian Schrittwieser, et al. 2024. Gemini 1.5: Un-	731
679	Zhang, Sheng Shen, and Yong Jae Lee. 2024a. Llava-	locking multimodal understanding across millions of	732
	next: Improved reasoning, ocr, and world knowledge .	tokens of context. <i>arXiv preprint arXiv:2403.05530</i> .	733
680			
681	Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae	Timo Schick, Jane Dwivedi-Yu, Roberto Dessi, Roberta	734
	Lee. 2023b. Visual instruction tuning.	Raileanu, Maria Lomeli, Eric Hambro, Luke Zettle-	735
682		moyer, Nicola Cancedda, and Thomas Scialom. 2024.	736
683	Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae	Toolformer: Language models can teach themselves	737
684	Lee. 2024b. Visual instruction tuning. <i>Advances in</i>	to use tools. <i>Advances in Neural Information Pro-</i>	738
	<i>neural information processing systems</i> , 36.	<i>cessing Systems</i> , 36.	739
685			
686	Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao	Dustin Schwenk, Apoorv Khandelwal, Christopher	740
687	Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang	Clark, Kenneth Marino, and Roozbeh Mottaghi. 2022.	741
688	Su, Jun Zhu, et al. 2023c. Grounding dino: Marrying	A-okvqa: A benchmark for visual question answer-	742
689	dino with grounded pre-training for open-set object	ing using world knowledge. In <i>European Conference</i>	743
	detection. <i>arXiv preprint arXiv:2303.05499</i> .	<i>on Computer Vision</i> , pages 146–162. Springer.	744
690			
691	Pan Lu, Swaroop Mishra, Tanglin Xia, Liang Qiu, Kai-	Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li,	745
692	Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter	Weiming Lu, and Yueting Zhuang. 2024. Hugging-	746
693	Clark, and Ashwin Kalyan. 2022. Learn to explain:	gpt: Solving ai tasks with chatgpt and its friends	747
694	Multimodal reasoning via thought chains for science	in hugging face. <i>Advances in Neural Information</i>	748
695	question answering. <i>Advances in Neural Information</i>	<i>Processing Systems</i> , 36.	749
696			
697	Anand Mishra, Shashank Shekhar, Ajeet Kumar Singh,	Amanpreet Singh, Vivek Natarajan, Meet Shah,	750
698	and Anirban Chakraborty. 2019. Ocr-vqa: Visual	Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh,	751
699	question answering by reading text in images. In	and Marcus Rohrbach. 2019. Towards vqa models	752
700	<i>2019 International Conference on Document Analy-</i>	that can read. In <i>Proceedings of the IEEE/CVF con-</i>	753
701	<i>sis and Recognition (ICDAR)</i> , pages 947–952. IEEE	<i>ference on computer vision and pattern recognition</i> ,	754
	Computer Society.	pages 8317–8326.	755
702			
703	Aitor Ormazabal, Che Zheng, Cyprien de Masson	Gemini Team, Rohan Anil, Sebastian Borgeaud,	756
704	d’Autume, Dani Yogatama, Deyu Fu, Donovan Ong,	Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu,	757
705	Eric Chen, Eugenie Lamprecht, Hai Pham, Isaac	Radu Soricut, Johan Schalkwyk, Andrew M Dai,	758
706	Ong, et al. 2024. Reka core, flash, and edge: A se-	Anja Hauth, et al. 2023. Gemini: a family of	759
707	ries of powerful multimodal language models. <i>arXiv</i>	highly capable multimodal models. <i>arXiv preprint</i>	760
	<i>preprint arXiv:2404.12387</i> .	<i>arXiv:2312.11805</i> .	761
708			
709	Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao,	Weihan Wang, Qingsong Lv, Wenmeng Yu, Wenyi	762
710	Shaohan Huang, Shuming Ma, and Furu Wei.	Hong, Ji Qi, Yan Wang, Junhui Ji, Zhuoyi Yang,	763
711	2023. Kosmos-2: Grounding multimodal large	Lei Zhao, Xixuan Song, et al. 2023a. Cogvlm: Vi-	764
712	language models to the world. <i>arXiv preprint</i>	sual expert for pretrained language models. <i>arXiv</i>	765
	<i>arXiv:2306.14824</i> .	<i>preprint arXiv:2311.03079</i> .	766
713			
714	Ji Qi, Ming Ding, Weihan Wang, Yushi Bai, Qingsong	Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le,	767
715	Lv, Wenyi Hong, Bin Xu, Lei Hou, Juanzi Li, Yux-	Ed H Chi, Sharan Narang, Aakanksha Chowdhery,	768
716	iao Dong, et al. 2024. Cogcom: Train large vision-	and Denny Zhou. 2022. Self-consistency improves	769
717	language models diving into details through chain of	chain of thought reasoning in language models. In	770
	manipulations. <i>arXiv preprint arXiv:2402.04236</i> .	<i>The Eleventh International Conference on Learning</i>	771
		<i>Representations</i> .	772

773	Yiqi Wang, Wentao Chen, Xiaotian Han, Xudong Lin,	Peter Young, Alice Lai, Micah Hodosh, and Julia Hock-	829
774	Haiteng Zhao, Yongfei Liu, Bohan Zhai, Jianbo Yuan,	enmaier. 2014. From image descriptions to visual	830
775	Quanzeng You, and Hongxia Yang. 2024. Exploring	denotations: New similarity metrics for semantic in-	831
776	the reasoning abilities of multimodal large language	ference over event descriptions. <i>Transactions of the</i>	832
777	models (mllms): A comprehensive survey on emerg-	<i>Association for Computational Linguistics</i> , 2:67–78.	833
778	ing trends in multimodal reasoning. <i>arXiv preprint</i>		
779	<i>arXiv:2401.06805</i> .		
780	Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa	Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng,	834
781	Liu, Noah A Smith, Daniel Khashabi, and Hannaneh	Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang,	835
782	Hajishirzi. 2023b. Self-instruct: Aligning language	Weiming Ren, Yuxuan Sun, et al. 2024. Mmmu: A	836
783	models with self-generated instructions. In <i>Proceed-</i>	massive multi-discipline multimodal understanding	837
784	<i>ings of the 61st Annual Meeting of the Association for</i>	and reasoning benchmark for expert agi. In <i>Pro-</i>	838
785	<i>Computational Linguistics (Volume 1: Long Papers)</i> ,	<i>ceedings of the IEEE/CVF Conference on Computer</i>	839
786	pages 13484–13508.	<i>Vision and Pattern Recognition</i> , pages 9556–9567.	840
787	Zilong Wang, Hao Zhang, Chun-Liang Li, Julian Mar-	Rowan Zellers, Yonatan Bisk, Ali Farhadi, and Yejin	841
788	tin Eisenschlos, Vincent Perot, Zifeng Wang, Lesly	Choi. 2019. From recognition to cognition: Vi-	842
789	Miculicich, Yasuhisa Fujii, Jingbo Shang, Chen-Yu	sual commonsense reasoning. In <i>Proceedings of the</i>	843
790	Lee, et al. 2023c. Chain-of-table: Evolving tables	<i>IEEE/CVF conference on computer vision and pat-</i>	844
791	in the reasoning chain for table understanding. In	<i>tern recognition</i> , pages 6720–6731.	845
792	<i>The Twelfth International Conference on Learning</i>		
793	<i>Representations</i> .	Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei,	846
794	Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel,	Nathan Scales, Xuezhi Wang, Dale Schuurmans,	847
795	Barret Zoph, Sebastian Borgeaud, Dani Yogatama,	Claire Cui, Olivier Bousquet, Quoc V Le, et al. 2022.	848
796	Maarten Bosma, Denny Zhou, Donald Metzler, et al.	Least-to-most prompting enables complex reasoning	849
797	2022a. Emergent abilities of large language models.	in large language models. In <i>The Eleventh Interna-</i>	850
798	<i>Transactions on Machine Learning Research</i> .	<i>tional Conference on Learning Representations</i> .	851
799	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten	Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang	852
800	Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou,	Wang, and Jifeng Dai. 2020. Deformable detr: De-	853
801	et al. 2022b. Chain-of-thought prompting elicits rea-	formable transformers for end-to-end object detec-	854
802	soning in large language models. <i>Advances in neural</i>	tion. <i>arXiv preprint arXiv:2010.04159</i> .	855
803	<i>information processing systems</i> , 35:24824–24837.		
804	Yang Wu, Shilong Wang, Hao Yang, Tian Zheng,	A Details of Data Construction.	856
805	Hongbo Zhang, Yanyan Zhao, and Bing Qin. 2023.		
806	An early evaluation of gpt-4v (ision). <i>arXiv preprint</i>	Attributes of Single-Entity Node. Each single-	857
807	<i>arXiv:2310.16534</i> .	entity node consists of five attributes: <i>Label</i> , <i>Lo-</i>	858
808	Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin	<i>cation</i> , <i>Color</i> , <i>Text</i> , and <i>Size</i> . Specifically, <i>Label</i>	859
809	Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu,	and <i>Location</i> are directly derived from the recogni-	860
810	Ce Liu, Michael Zeng, and Lijuan Wang. 2023. Mm-	tion results of Deformable DETR. <i>Color</i> is deter-	861
811	react: Prompting chatgpt for multimodal reasoning	mined using ColorThief ¹⁴ to analyze the dominant	862
812	and action. <i>arXiv preprint arXiv:2303.11381</i> .	color scheme of the subgraph corresponding to the	863
813	Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran,	single-entity node. <i>Text</i> is obtained by applying	864
814	Tom Griffiths, Yuan Cao, and Karthik Narasimhan.	PaddleOCR to recognize textual content within the	865
815	2024. Tree of thoughts: Deliberate problem solving	image. <i>Size</i> records the node’s proportion in the	866
816	with large language models. <i>Advances in Neural</i>	entire image, considering its width, height, and	867
817	<i>Information Processing Systems</i> , 36.	area.	868
818	Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak	Construction of Chain. We use tools that pro-	869
819	Shafran, Karthik R Narasimhan, and Yuan Cao. 2022.	duce graphical outputs (e.g., Grounding, Highlight)	870
820	React: Synergizing reasoning and acting in language	to connect intermediary nodes and tools that pro-	871
821	models. In <i>The Eleventh International Conference</i>	duce textual outputs (e.g., OCR, Answer) to con-	872
822	<i>on Learning Representations</i> .	nect terminal nodes.	873
823	Qinghao Ye, Haiyang Xu, Guohai Xu, Jiabo Ye,	We construct the chain by sequentially adding	874
824	Ming Yan, Yiyang Zhou, Junyang Wang, An-	nodes. Specifically, we maintain a queue $L =$	875
825	wen Hu, Pengcheng Shi, Yaya Shi, et al. 2023.	$(N_1, N_2, \dots, N_i)$ consisting of nodes. To add	876
826	mplug-owl: Modularization empowers large lan-	N_{i+1} , we first specify the tool t_i to be used and	877
827	guage models with multimodality. <i>arXiv preprint</i>		
828	<i>arXiv:2304.14178</i> .		

¹⁴<https://lokeshdhakar.com/projects/color-thief/>

sample a node in the remaining set $\hat{N} = \{N_j \notin L\}$ that allow the use of this tool. For instance, only nodes with text in images permit using the OCR tool. If $\hat{N} = \emptyset$ or the chain length reaches the limit, the process terminates. When constructing VIREO, we set the maximum chain length to 4. Consequently, the dataset includes reasoning paths with lengths of 2, 3, and 4.

B Implementation Details

Image Source. We use the COCO2014 dataset (Lin et al., 2014) as our source of images.

Questioner. We use LLaMA-3-8B-Instruct as the base model for the Questioner. To construct the corresponding training data, we first manually create 5 seed prompts for each combination of head node and tail node ($3 \times 3 = 9$). Then, we generate a total of 10k instances by using GPT-4. During this process, to reduce the bias introduced by the seed prompts, we gradually add the obtained results into the prompt pool. For each instance, three prompts are randomly selected from the prompt pool as demonstrations. For each instance, the input contains 3 fields: the profiles of the head node and tail node, and the specified tool. Output includes 2 fields: the question and the arguments of the tool. We perform instruction fine-tuning on the Questioner using LoRA. The rank is set to 8 and the lora_alpha is set to 8. We adopt AdamW as the optimizer. We set adam_beta1, adam_beta2, and adam_epsilon to 0.9, 0.999, and 1e-8, respectively. We use the cosine schedule to warm up the training and set warmup_steps to 0.1. We set the batch size to 4 and fine-tune Questioner for 2 epochs, with each epoch taking around 1 hour. The training process is completed with 8 Nvidia A100 GPUs.

Combiner. Similar to Questioner, we also use LLaMA-3-8B-Instruct as the base model for the Combiner. We manually create 20 seed prompts and generate a total of 10k instances by using GPT-4. The input includes two questions to be merged, and the output is the merged result. The fine-tuning settings of Combiner keep the same as Questioner.

Reasoner. We synthesize 50k data examples using the least-to-most synthesis method. Each example includes the current image I , the main question Q , and the previous sub-questions $\{q_{<k}\}$. The label for each data example comprises the current

sub-question q_k and the tool t_k that needs to be invoked. We use this data to train LLaVA-1.5-7B as the Reasoner. We perform instruction fine-tuning on the Reasoner using LoRA. The rank is set to 8 and the lora_alpha is set to 8. We adopt AdamW as the optimizer. We set adam_beta1, adam_beta2, and adam_epsilon to 0.9, 0.999, and 1e-8, respectively. We use the cosine schedule to warm up the training and set warmup_steps to 0.1. We set the batch size to 8 and fine-tune the Reasoner for 3 epochs, with each epoch taking around 3 hours. The training process is completed with 1 Nvidia A100 GPU.

C Case Study

To facilitate understanding of our method, we show some cases in Figure 5, Figure 6, and Figure 7.

Question: What's written on the top of the donut? 	 <i>Sub Question: Where is the donut?</i> <i>Tool: Grounding</i> <i>Arguments: donut</i>
	 <i>Sub Question: What is written on the top?</i> <i>Tool: Grounding</i> <i>Arguments: Text on the top</i>
	 <i>Sub Question: What is written on this?</i> <i>Tool: OCR</i>
	 <i>Tool: Answer</i> <i>Arguments: What is written on this?</i> <i>Character: RANDY's</i>
	Answer: RANDY' s

Figure 5: Case 1. This case uses Grounding to locate the donut and uses OCR and Answer to get the final answer.

Question: What color is the man in orange's bicycle? 	 <i>Sub Question: Where is the man in orange?</i> <i>Tool: Grounding</i> <i>Arguments: man in orange</i>
	 <i>Sub Question: Where is the bicycle near the man?</i> <i>Tool: Grounding</i> <i>Arguments: bicycle near the man</i>
	 <i>Sub Question: What is the color of this bicycle?</i> <i>Tool: Answer</i>
	Answer: The bicycle is white.

Figure 6: Case 2. In this case, the man wearing orange is first precisely identified, and then attention is directed to the bicycle near him to obtain the answer.

Question: How many people are rowing? 	 Sub Question: <i>Where are the people rowing?</i> Tool: <i>Grounding</i> Arguments: <i>people rowing a boat</i>  Sub Question: <i>How many people are rowing the boat?</i> Tool: <i>Highlight</i> Arguments: <i>people rowing a boat</i>  Sub Question: <i>How many people are rowing the boat?</i> Tool: <i>Answer</i> Answer: 8
---	---

Figure 7: Case 3. This case excludes irrelevant people and highlights the target group, thereby achieving accurate counting results.